

CIT 4299: IT Project
CIT 4299: Systems Project

Proposal Writing guide (June 2026 Edition)

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Note on AI use (read carefully)

You may use AI tools for limited purposes:

- Brainstorming topic ideas
- Checking grammar and spelling
- Suggesting better sentence structure
- Reviewing whether a section is unclear

You may NOT use AI to:

- Write complete sentences or paragraphs for you (even if you later rephrase them)
- Generate your problem statement, objectives, or literature review summaries
- Produce your architecture diagram or methodology description
- Create citations or fake references

Important: If you copy-paste AI-generated content, or write something and then ask AI to ‘improve it’ and paste that back, this is considered as cheating in a University examination, and you will face the appropriate disciplinary procedure. Proposals will be screened using AI detection tools. Flagged submissions will trigger an oral defense before any mark is awarded.

The core test: Can you explain every section of this proposal in your own words during a 10-minute Q&A with your supervisor? If not, you have used AI inappropriately.

Use proper citations for all external references, tools, and content — including any AI-generated material that influenced your thinking (cite the tool and date).

Proposal Structure: 10 chapters

Your proposal should follow the structure below. Keep the total length between 8–12 pages (excluding references and diagrams).

1. Introduction (under 1 page)

Give a general overview of the domain or area you are exploring.

- Set the context: Why is this domain relevant today (2026)?
- Identify the gap or need you want to address.
- Avoid buzzwords and hype; focus on clarity.
- Do not let AI write this for you. Think of it as explaining your project to a friend in a different field.

2. Problem statement

Clearly define the real-world problem your project addresses.

- State what the problem is.
- Who is affected? Why is it important to solve it?
- Use evidence: a statistic, observation, or citation from literature or industry.

- Avoid vague or generic problems.

Example: “Smallholder farmers in Kenya lack timely weather-based crop advisories, leading to poor planting decisions and 20–30% yield losses during unexpected rains (Source: Kenya Agricultural Institute, 2024).”

3. Objectives

Define what your project aims to achieve.

- Use SMART criteria (Specific, Measurable, Achievable, Relevant, Time-bound).
- List 3–5 objectives.
- Distinguish between primary (must-have) and secondary (nice-to-have) objectives.

Weak: “Build a website to ...”

Improved:

By the end of November 2026, I will: i. Design a responsive web platform for real-time weather alerts tailored for rural farmers. ii. Implement the design in (i) using React and Firebase, with SMS fallback for low-connectivity areas.”

4. Literature Review

Show you have studied existing work in your area.

- Summarize at least 8–12 relevant sources (academic papers, industry reports, technical documentation, or standards).
- Group findings thematically (e.g., techniques, success factors, limitations, existing systems).
- Identify gaps or limitations that your project will address.
- Properly cite all references using APA, IEEE, or Harvard style.
- Do not copy AI-generated summaries. Read the source, then write in your own words.

Where to find sources: Google Scholar, IEEE Xplore, ACM Digital Library, ResearchGate, arXiv.

5. Proposed system architecture

Present a high-level design of your system.

- Describe components/modules (backend, frontend, database, APIs, ML models, etc.).
- Include a simple architecture diagram (block diagram or flowchart) - use draw.io, Lucidchart, or even a clear hand-drawn photo.
- Show how components interact and how data flows between them.
- Explain your technology choices (frameworks, libraries, platforms) with justification.
- State your target deployment environment (e.g., Vercel, Railway, AWS free tier, local Docker, or a Raspberry Pi).
- Mention whether you will use CI/CD (e.g., GitHub Actions, GitLab CI) - even a simple ‘push to main triggers deploy’ is acceptable.

6. Methodology

Describe your development and research approach.

- State your development model (Agile/Scrum, Waterfall, or a hybrid).

- Describe phases: requirements gathering, design, development, testing, deployment. These must reflect your project, not just picking from the generalized class notes.
- State your testing strategy:
 - Unit tests (name the framework: Jest, PyTest, JUnit, etc.)
 - Integration tests
 - User acceptance testing (if applicable)
- For AI/ML projects: state which model(s) you plan to use, how you will train/test, and how you will measure accuracy, latency, or cost per inference.
- State how you will track progress (e.g., weekly GitHub commits, a simple Kanban board, or a sprint log).

Success Criteria:

List 3–5 measurable outcomes that will tell you the project succeeded.

Examples:

- “System returns a weather advisory within 2 seconds for 95% of requests.”
- “A farmer unfamiliar with the system can request and receive an alert in under 3 minutes without training.”
- “The system runs within free-tier limits of chosen cloud platform for the first 6 months.”

7. Timeline

Break your project into weekly deliverables.

- Use a Gantt chart or table format (Weeks 1–25, or across both semesters).
- Link activities to milestones (proposal submission, MVP, integration, testing, final system, report draft, final submission).
- Highlight critical deadlines.

You can use: Spreadsheets (eg Google Sheets), GanttProject, or a simple markdown table.

8. Budget

Estimate the cost of completing your project.

Include items such as:

- Hosting fees (e.g., Vercel, Railway, AWS, Firebase)
- API or AI model usage costs (OpenAI, Together AI, Google Gemini, Anthropic, etc.)
- Cloud computing time (GPU instances if training models)
- Devices or sensors (if IoT/hardware project)
- Data collection costs (transport, SIM cards, survey printing)
- Domain name (if needed — optional)

If your project can be completed with zero monetary cost, state exactly how (e.g., “All hosting on free tiers: GitHub Pages for frontend, Supabase free tier for database, Hugging Face Inference API free for model.”)

Do not write “N/A” - think through every resource you will touch.

9. Ethical considerations

Demonstrate responsibility in your project.

Discuss:

- Data privacy: If you collect user data, how will you store it securely? Will you anonymize it?
- Bias and fairness: For AI projects, how will you check that your model does not discriminate?
- Informed consent: If testing with real users, how will you obtain permission?
- Accessibility: Will your system work for users with disabilities or low-bandwidth connections?
- Open source and code reuse: If you use existing code, libraries, or templates, list them here and state how you will comply with their licenses (MIT, GPL, Apache, etc.).

A proposal that does not show responsibilities will be rejected.

10. References

Acknowledge all sources you used.

- Use a consistent citation style (APA, Harvard, or IEEE).
- Include scholarly sources, websites, official documentation for libraries/tools, and datasets.
- If you used an AI tool for brainstorming or grammar checking, cite it as:

Example (APA):

OpenAI. (2026). ChatGPT (Mar 2026 version) [Large language model]. <https://chat.openai.com/>

- If you did not use AI for any part of the proposal, you may state: “No AI tools were used in the preparation of this proposal.”

Success Checklist (before you submit)

Item	Done?
All 10 chapters complete	☐
At least 8 sources in Literature Review	☐
Architecture diagram included	☐
Deployment environment stated	☐
Testing strategy described (with framework names)	☐
Success Criteria section present (3–5 measurable outcomes)	☐
Budget includes API/AI costs if applicable	☐
Ethical considerations include data privacy and open source disclosure	☐
You can explain every section in your own words	☐
Proposal is entirely your own writing (no AI-generated sentences)	☐

What happens after submission

- Your proposal will be reviewed by your supervisor.
- Proposals will be screened with AI detection tools.
- You will have a 10–15 minute oral defense where you explain and justify your proposal without slides.
- If you cannot explain a section in your own words, your proposal will be returned for revision or rejected.

Misuse of AI (copy-paste, rephrased AI content, fake references) may lead to proposal rejection and a formal refer to academic misconduct committee.

Final advice

A good proposal is one you fully understand and believe in. Your supervisor is not your enemy — they want you to succeed. If you are stuck, ask them for help before you ask AI to write your problem statement, for example.

Good luck.

- The unit coordinators